

Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with @BENCH adt_fields; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	ADT fields	SoA bufs	Fold passes			Map passes		
			AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
Compiler	9	8	0.1233	0.0391	3.15 ×	0.1575	0.1778	0.89×
DBQuery	15	13	0.1176	0.0879	1.34 ×	0.1395	0.1653	0.84×
DecisionTree	7	6	3.640	2.959	1.23 ×	—	—	—
DomTree	15	14	0.1350	0.0605	2.23 ×	0.0579	0.0789	0.73×
KDTree	17	16	0.1742	0.1386	1.26 ×	—	—	—
LinearListReduction	11	11	0.0290	0.0050	5.84 ×	—	—	—
List	2	2	0.0882	0.0895	0.99×	0.2927	0.3585	0.82×
MonoTree	3	2	0.0170	0.0225	0.76×	0.0453	0.0427	1.06×
ObjectGraph	5	4	0.0880	0.0792	1.11 ×	0.1427	0.1620	0.88×
OctTree	16	9	0.0356	0.0310	1.15 ×	0.0378	0.0440	0.86×
PiecewiseFunctions	8	7	0.1346	0.0910	1.48 ×	0.2387	0.2577	0.93×
TernaryTree	5	3	0.0278	0.0262	1.06×	0.0831	0.0778	1.07×
Trie	10	9	0.0634	0.0318	1.99 ×	0.1488	0.1657	0.90×
ColorOctree	25	18	0.0958	0.0767	1.25 ×	—	—	—

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	CYC	INS	L1D	L1I	L2D	L2I	LLC
Compiler	1.57 ×	0.52×	4.22 ×	0.79×	14.09 ×	0.32×	6.80 ×
DBQuery	1.05×	0.47×	1.34 ×	0.58×	8.44 ×	0.13×	6.33 ×
DecisionTree	1.23 ×	0.59×	17.90 ×	2.36 ×	0.84×	2.50 ×	37.09 ×
DomTree	1.52 ×	0.57×	3.08 ×	0.98×	21.91 ×	0.22×	10.49 ×
KDTree	1.26 ×	0.77×	1.28 ×	0.72×	0.25×	0.74×	4.86 ×
LinearListReduction	5.87 ×	0.90×	32.86 ×	7.54 ×	40.49 ×	7.43 ×	11.78 ×
List	0.82×	0.80×	3.85 ×	0.51×	0.09×	0.81×	1.45 ×
MonoTree	0.96×	0.85×	0.52×	0.88×	0.23×	0.68×	1.23 ×
ObjectGraph	0.94×	0.75×	3.95 ×	0.90×	0.55×	0.44×	2.93 ×
OctTree	1.01×	0.73×	0.58×	0.46×	0.19×	0.65×	3.79 ×
PiecewiseFunctions	1.12 ×	0.60×	2.87 ×	0.64×	2.18 ×	0.51×	4.48 ×
TernaryTree	1.13 ×	0.79×	0.32×	0.48×	0.22×	1.22 ×	1.40 ×
Trie	1.13 ×	0.56×	3.36 ×	0.85×	3.29 ×	0.12×	6.35 ×
ColorOctree	1.25 ×	0.84×	3.12 ×	1.35 ×	0.38×	1.36 ×	4.58 ×
Geomean	1.28 ×	0.68×	2.58 ×	0.93×	1.27 ×	0.64×	4.80 ×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Compiler	9.3e+08/ 5.9e+08	9.9e+08 /1.9e+09	1.3e+07/ 3.1e+06	2.1e+05 /2.6e+05	1.2e+06/ 8.6e+04	1.0e+04 /3.2e+04	5.1e+07/ 7.5e+06
DBQuery	9.3e+08/ 8.8e+08	1.1e+09 /2.2e+09	3.9e+06/ 2.9e+06	1.9e+05 /3.2e+05	8.1e+05/ 9.6e+04	4.1e+03 /3.2e+04	2.1e+07/ 3.3e+06
DecisionTree	1.8e+10/ 1.5e+10	4.0e+10 /6.8e+10	5.4e+08/ 3.0e+07	1.1e+05/ 4.7e+04	1.8e+06 /2.1e+06	1.1e+05/ 4.4e+04	1.1e+09/ 2.9e+07
DomTree	8.1e+08/ 5.3e+08	1.2e+09 /2.2e+09	1.2e+07/ 3.9e+06	1.3e+05 /1.3e+05	1.1e+06/ 4.9e+04	6.9e+03 /3.2e+04	5.7e+07/ 5.4e+06
KDTree	8.8e+08/ 7.0e+08	3.1e+09 /4.0e+09	1.4e+07/ 1.1e+07	7.6e+02 /1.0e+03	2.3e+04 /9.3e+04	7.7e+02 /1.0e+03	4.4e+07/ 9.1e+06
LinearListReduction	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	3.5e+02/ 4.6e+01	2.8e+05/ 6.9e+03	3.5e+02/ 4.7e+01	1.2e+07/ 1.0e+06
List	1.2e+09 /1.5e+09	4.5e+09 /5.6e+09	9.4e+06/ 2.4e+06	2.9e+05 /5.7e+05	1.9e+04 /2.0e+05	1.1e+04 /1.4e+04	4.8e+07/ 3.3e+07
MonoTree	2.5e+08 /2.6e+08	9.1e+08 /1.1e+09	2.4e+05 /4.7e+05	4.0e+04 /4.5e+04	2.9e+03 /1.3e+04	1.3e+03 /1.9e+03	2.0e+06/ 1.7e+06
ObjectGraph	8.2e+08 /8.7e+08	3.0e+09 /4.0e+09	9.0e+06/ 2.3e+06	2.0e+05 /2.2e+05	2.9e+04 /5.3e+04	7.0e+03 /1.6e+04	2.3e+07/ 8.0e+06
OctTree	2.7e+08/ 2.7e+08	8.2e+08 /1.1e+09	6.4e+05 /1.1e+06	5.2e+04 /1.1e+05	4.8e+03 /2.5e+04	2.5e+03 /3.8e+03	4.3e+06/ 1.1e+06
PiecewiseFunctions	1.2e+09/ 1.1e+09	2.8e+09 /4.7e+09	1.5e+07/ 5.1e+06	3.1e+05 /4.9e+05	2.0e+05/ 9.0e+04	2.8e+04 /5.5e+04	4.5e+07/ 1.0e+07
TernaryTree	4.1e+08/ 3.6e+08	1.2e+09 /1.5e+09	5.2e+05 /1.6e+06	6.8e+04 /1.4e+05	4.4e+03 /2.0e+04	2.9e+03/ 2.4e+03	3.3e+06/ 2.4e+06
Trie	6.5e+08/ 5.7e+08	1.2e+09 /2.2e+09	8.9e+06/ 2.7e+06	1.8e+05 /2.2e+05	1.7e+05/ 5.0e+04	4.9e+03 /4.0e+04	2.6e+07/ 4.0e+06
ColorOctree	4.9e+08/ 3.9e+08	1.6e+09 /2.0e+09	9.6e+06/ 3.1e+06	5.5e+02/ 4.1e+02	2.7e+04 /7.3e+04	5.6e+02/ 4.1e+02	1.8e+07/ 4.0e+06

Table 4: Mutable vs immutable cursor comparison (times only). Times are median per iteration (s). **Bold** marks the fastest time across all four variants.

Program	Am	Ai	Sm	Si
Compiler	0.2807	1.083	0.2170	<i>OOM</i>
DBQuery	0.2571	0.3229	0.2532	1.118
DecisionTree	3.640	5.583	2.959	24.81
DomTree	0.1929	0.3336	0.1395	0.5277
KDTree	0.1742	0.2419	0.1386	0.7211
LinearListReduction	0.0290	0.1323	0.0050	<i>OOM</i>
List	0.3809	<i>OOM</i>	0.4480	<i>OOM</i>
MonoTree	0.0623	0.1411	0.0651	0.2737
ObjectGraph	0.2308	0.4840	0.2413	0.7065
OctTree	0.0734	0.0811	0.0750	<i>OOM</i>
PiecewiseFunctions	0.3734	0.5558	0.3487	2.164
TernaryTree	0.1109	0.1437	0.1040	0.6159
Trie	0.2123	0.2915	0.1975	1.005
ColorOctree	0.0958	0.1160	0.0767	<i>OOM</i>

Table 5: Mutable vs immutable cursor comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut, AoS-imm/SoA-imm. >1× means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Compiler	1.29 ×	3.86 ×	4.99 ×	—
DBQuery	1.02×	1.26 ×	1.28 ×	0.29×
DecisionTree	1.23 ×	1.53 ×	1.89 ×	0.22×
DomTree	1.38 ×	1.73 ×	2.39 ×	0.63×
KDTree	1.26 ×	1.39 ×	1.74 ×	0.34×
LinearListReduction	5.84 ×	4.56 ×	26.64 ×	—
List	0.85×	—	—	—
MonoTree	0.96×	2.26 ×	2.17 ×	0.52×
ObjectGraph	0.96×	2.10 ×	2.01 ×	0.69×
OctTree	0.98×	1.11 ×	1.08×	—
PiecewiseFunctions	1.07×	1.49 ×	1.59 ×	0.26×
TernaryTree	1.07×	1.30 ×	1.38 ×	0.23×
Trie	1.07×	1.37 ×	1.48 ×	0.29×
ColorOctree	1.25 ×	1.21 ×	1.51 ×	—

Table 6: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
blockCountPass	F	2/9	78%	0.0116 $\pm$ 0.00001	0.0793 $\pm$ 0.00002	0.0019$\pm$0.00000	<i>OOM</i>	6.06$\times$	6.82$\times$	41.37$\times$	–
branchStatsPass	F	2/9	78%	0.0114 $\pm$ 0.00001	0.0794 $\pm$ 0.00002	0.0036$\pm$0.00000	<i>OOM</i>	3.20$\times$	6.98$\times$	22.34$\times$	–
castInstCountPass	F	3/9	67%	0.0177 $\pm$ 0.00001	0.0794 $\pm$ 0.00087	0.0013$\pm$0.00000	<i>OOM</i>	13.71$\times$	4.49$\times$	61.52$\times$	–
has cycle	F	4/9	56%	0.0239 $\pm$ 0.00010	0.0928 $\pm$ 0.00116	0.0173$\pm$0.00015	<i>OOM</i>	1.38$\times$	3.88$\times$	5.36$\times$	–
instCountPass	F	2/9	78%	0.0117 $\pm$ 0.00037	0.0796 $\pm$ 0.00154	0.0019$\pm$0.00000	<i>OOM</i>	6.02$\times$	6.79$\times$	40.83$\times$	–
latencyModelPass	F	3/9	67%	0.0116 $\pm$ 0.00001	0.0799 $\pm$ 0.00005	0.0030$\pm$0.00000	<i>OOM</i>	3.93$\times$	6.86$\times$	27.01$\times$	–
memoryOpStatsPass	F	3/9	67%	0.0115 $\pm$ 0.00001	0.0794 $\pm$ 0.00119	0.0038$\pm$0.00001	<i>OOM</i>	3.02$\times$	6.93$\times$	20.91$\times$	–
stripSideEffectsPass	M	7/9	22%	0.0790$\pm$0.00194	0.1831 $\pm$ 0.00018	0.0896 $\pm$ 0.00130	<i>OOM</i>	0.88 $\times$	2.32$\times$	2.04$\times$	–
targetReturnPass	M	9/9	0%	0.0784$\pm$0.00017	0.1813 $\pm$ 0.00112	0.0883 $\pm$ 0.00023	<i>OOM</i>	0.89 $\times$	2.31$\times$	2.05$\times$	–
throughputModelPass	F	3/9	67%	0.0117 $\pm$ 0.00001	0.0793 $\pm$ 0.00002	0.0030$\pm$0.00000	<i>OOM</i>	3.96$\times$	6.79$\times$	26.87$\times$	–
verifyIR	F	9/9	0%	0.0121 $\pm$ 0.00005	0.0701 $\pm$ 0.00006	0.0034$\pm$0.00005	<i>OOM</i>	3.55$\times$	5.77$\times$	20.49$\times$	–
Total				0.2807	1.083	0.2170	–	1.29$\times$	3.86$\times$	4.99$\times$	–
Geomean								3.14$\times$	5.06$\times$	15.88$\times$	–

Table 7: Per-pass PAPI counters for **Compiler** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
blockCountPass	F	5.9e+07/ 9.7e+06	5.1e+07/ 4.5e+07	1.2e+06/ 6.2e+03	1.6e+02/ 2.3e+01	2.2e+05/ 1.1e+03	1.6e+02/ 2.3e+01	4.7e+06/ 3.2e+02
branchStatsPass	F	5.8e+07/ 1.8e+07	7.9e+07 /8.6e+07	8.2e+05/ 2.1e+04	1.1e+02/ 4.7e+01	4.8e+04/ 3.2e+03	1.1e+02/ 4.9e+01	4.6e+06/ 4.3e+05
castInstCountPass	F	9.0e+07/ 6.6e+06	5.1e+07/ 3.9e+07	1.4e+04 /2.2e+04	1.1e+02/ 2.9e+01	9.2e+02 /1.1e+03	1.1e+02/ 2.9e+01	4.6e+06/ 3.3e+03
has cycle	F	1.2e+08/ 8.8e+07	1.1e+08 /1.5e+08	2.4e+06/ 2.6e+05	2.9e+02/ 1.4e+02	1.5e+05/ 2.2e+04	2.9e+02/ 1.4e+02	5.2e+06/ 2.3e+06
instCountPass	F	5.9e+07/ 9.9e+06	5.6e+07 /5.7e+07	1.2e+06/ 1.6e+04	4.0e+02/ 3.3e+01	2.2e+05/ 1.1e+03	4.0e+02/ 3.3e+01	4.7e+06/ 5.2e+02
latencyModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.4e+04	1.7e+02/ 4.3e+01	2.0e+05/ 3.9e+03	1.7e+02/ 4.3e+01	4.7e+06/ 5.2e+05
memoryOpStatsPass	F	5.8e+07/ 1.9e+07	8.5e+07 /9.1e+07	6.6e+05/ 3.1e+04	1.5e+02/ 4.4e+01	2.8e+04/ 3.2e+03	1.6e+02/ 4.4e+01	4.6e+06/ 4.4e+05
stripSideEffectsPass	M	1.5e+08 /2.0e+08	1.8e+08 /6.0e+08	1.8e+06/ 1.2e+06	1.1e+05 /1.4e+05	6.2e+03 /1.9e+04	4.6e+03 /1.2e+04	4.1e+06/ 1.2e+06
targetReturnPass	M	1.5e+08 /1.9e+08	1.7e+08 /6.3e+08	1.6e+06/ 1.4e+06	9.1e+04 /1.2e+05	5.7e+03 /2.4e+04	3.5e+03 /2.0e+04	4.1e+06/ 1.5e+06
throughputModelPass	F	5.9e+07/ 1.5e+07	6.2e+07 /6.9e+07	1.2e+06/ 3.1e+04	2.8e+02/ 4.5e+01	2.0e+05/ 3.5e+03	2.8e+02/ 4.5e+01	4.8e+06/ 5.4e+05
verifyIR	F	6.2e+07/ 1.7e+07	7.4e+07 /8.6e+07	8.9e+05/ 1.8e+04	4.7e+02/ 1.3e+02	1.3e+05/ 4.1e+03	4.6e+02/ 1.2e+02	4.7e+06/ 4.6e+05
Total		9.3e+08/5.9e+08	9.9e+08/1.9e+09	1.3e+07/3.1e+06	2.1e+05/2.6e+05	1.2e+06/8.6e+04	1.0e+04/3.2e+04	5.1e+07/7.5e+06

Table 8: Per-pass PAPI counters for **Compiler** (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
blockCountPass	F	4.0e+08/–	1.6e+08/–	2.2e+06/–	7.4e+02/–	9.1e+04/–	7.4e+02/–	1.4e+07/–
branchStatsPass	F	4.0e+08/–	1.8e+08/–	2.1e+06/–	6.8e+02/–	4.3e+04/–	6.8e+02/–	1.4e+07/–
castInstCountPass	F	4.0e+08/–	1.8e+08/–	2.1e+06/–	6.9e+02/–	4.5e+04/–	6.8e+02/–	1.4e+07/–
has cycle	F	4.7e+08/–	2.1e+08/–	2.5e+06/–	3.1e+02/–	7.0e+04/–	3.1e+02/–	1.7e+07/–
instCountPass	F	4.0e+08/–	1.6e+08/–	2.2e+06/–	5.8e+02/–	9.8e+04/–	5.8e+02/–	1.4e+07/–
latencyModelPass	F	4.0e+08/–	1.7e+08/–	2.1e+06/–	5.5e+02/–	4.4e+04/–	5.6e+02/–	1.4e+07/–
memoryOpStatsPass	F	4.0e+08/–	1.9e+08/–	2.0e+06/–	5.7e+02/–	4.2e+04/–	5.8e+02/–	1.4e+07/–
stripSideEffectsPass	M	6.7e+08/–	3.3e+08/–	1.8e+06/–	1.1e+05/–	1.7e+04/–	1.8e+04/–	2.3e+07/–
targetReturnPass	M	6.6e+08/–	3.2e+08/–	2.0e+06/–	1.0e+05/–	1.8e+04/–	2.4e+04/–	2.3e+07/–
throughputModelPass	F	4.0e+08/–	1.7e+08/–	2.1e+06/–	5.0e+02/–	4.4e+04/–	4.9e+02/–	1.4e+07/–
verifyIR	F	3.6e+08/–	1.2e+08/–	2.5e+06/–	3.8e+02/–	2.0e+05/–	3.8e+02/–	1.1e+07/–
Total		5.0e+09/–	2.2e+09/–	2.4e+07/–	2.2e+05/–	7.1e+05/–	4.7e+04/–	1.7e+08/–

Table 9: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
clearQueryFlags	M	14/15	7%	0.0699$\pm$0.00009	0.0755 $\pm$ 0.00009	0.0814 $\pm$ 0.00111	0.2998 $\pm$ 0.00214	0.86 $\times$	1.08 $\times$	0.93 $\times$	0.25 $\times$
countJoins	F	3/15	80%	0.0184 $\pm$ 0.00001	0.0279 $\pm$ 0.00001	0.0112$\pm$0.00076	0.0847 $\pm$ 0.00003	1.63$\times$	1.52$\times$	2.48$\times$	0.33 $\times$
filterSelectivitySkew	F	5/15	67%	0.0185 $\pm$ 0.00001	0.0276 $\pm$ 0.00012	0.0122$\pm$0.00001	0.0862 $\pm$ 0.00002	1.51$\times$	1.50$\times$	2.26$\times$	0.32 $\times$
hashJoinPressure	F	5/15	67%	0.0187 $\pm$ 0.00000	0.0319 $\pm$ 0.00002	0.0124$\pm$0.00109	0.0857 $\pm$ 0.00019	1.51$\times$	1.70$\times$	2.57$\times$	0.37 $\times$
scaleCosts	M	15/15	0%	0.0696$\pm$0.00011	0.0754 $\pm$ 0.00009	0.0839 $\pm$ 0.00006	0.2979 $\pm$ 0.00258	0.83 $\times$	1.08 $\times$	0.90 $\times$	0.25 $\times$
sumCost	F	6/15	60%	0.0184 $\pm$ 0.00036	0.0276 $\pm$ 0.00013	0.0148$\pm$0.00009	0.0880 $\pm$ 0.00003	1.24$\times$	1.50$\times$	1.86$\times$	0.31 $\times$
sumMemory	F	6/15	60%	0.0182 $\pm$ 0.00001	0.0279 $\pm$ 0.00001	0.0152$\pm$0.00002	0.0881 $\pm$ 0.00002	1.20$\times$	1.53$\times$	1.84$\times$	0.32 $\times$
sumRows	F	6/15	60%	0.0255 $\pm$ 0.00091	0.0292 $\pm$ 0.00002	0.0220$\pm$0.00082	0.0871 $\pm$ 0.00016	1.16$\times$	1.15$\times$	1.33$\times$	0.34 $\times$
Total				0.2571	0.3229	0.2532	1.118	1.02 $\times$	1.26$\times$	1.28$\times$	0.29 $\times$
Geomean								1.21$\times$	1.36$\times$	1.65$\times$	0.31 $\times$

Table 10: Per-pass PAPI counters for **DBQuery** (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
clearQueryFlags	M	1.7e+08 /2.2e+08	1.9e+08 /7.2e+08	3.1e+05 /1.2e+06	9.7e+04 /1.6e+05	4.1e+04/ 2.3e+04	1.1e+03 /1.4e+04	2.5e+06/ 9.2e+05
countJoins	F	9.3e+07/ 5.7e+07	7.4e+07/ 7.1e+07	6.1e+05/ 2.5e+04	4.1e+02/ 9.8e+01	1.4e+05/ 2.0e+03	4.1e+02/ 5.9e+01	2.6e+06/ 9.7e+02
filterSelectivitySkew	F	9.3e+07/ 6.2e+07	1.0e+08 /1.1e+08	5.2e+05/ 7.5e+04	2.4e+02/ 8.1e+01	1.2e+05/ 6.2e+03	2.4e+02/ 8.2e+01	2.7e+06/ 2.3e+05
hashJoinPressure	F	9.5e+07/ 6.3e+07	8.4e+07 /9.9e+07	7.3e+05/ 9.6e+04	3.5e+02/ 1.0e+02	1.9e+05/ 7.4e+03	3.5e+02/ 1.0e+02	2.7e+06/ 1.2e+05
scaleCosts	M	1.7e+08 /2.2e+08	2.2e+08 /7.5e+08	2.9e+05 /1.3e+06	8.7e+04 /1.6e+05	3.6e+04/ 2.8e+04	1.4e+03 /1.8e+04	2.5e+06/ 1.1e+06
sumCost	F	9.3e+07/ 7.5e+07	9.0e+07 /1.2e+08	5.5e+05/ 7.8e+04	1.2e+02/ 5.9e+01	1.2e+05/ 1.1e+04	1.2e+02/ 6.0e+01	2.7e+06/ 3.2e+05
sumMemory	F	9.2e+07/ 7.7e+07	9.0e+07 /1.2e+08	6.7e+05/ 7.9e+04	3.3e+02/ 5.6e+01	1.6e+05/ 1.1e+04	3.2e+02/ 5.6e+01	2.7e+06/ 3.1e+05
sumRows	F	1.3e+08/ 1.1e+08	2.1e+08 /2.3e+08	2.8e+05/ 1.1e+05	1.2e+02/ 1.0e+02	2.2e+04/ 7.6e+03	1.2e+02/ 1.1e+02	2.4e+06/ 2.8e+05
Total		9.3e+08/8.8e+08	1.1e+09/2.2e+09	3.9e+06/2.9e+06	1.9e+05/3.2e+05	8.1e+05/9.6e+04	4.1e+03/3.2e+04	2.1e+07/3.3e+06

Table 11: Per-pass PAPI counters for DBQuery (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
clearQueryFlags	M	2.0e+08 /1.3e+09	3.2e+08 /2.4e+09	5.7e+05 /2.0e+07	9.2e+04 /6.7e+05	1.9e+04 /4.7e+04	2.2e+03 /4.0e+04	2.3e+06/ 9.9e+05
countJoins	F	1.4e+08 /4.3e+08	1.6e+08 /7.7e+08	1.9e+06/ 3.2e+05	1.8e+02 /6.5e+02	1.1e+05/ 1.3e+04	1.7e+02/ 1.0e+02	2.3e+06/ 2.3e+04
filterSelectivitySkew	F	1.4e+08 /4.4e+08	2.0e+08 /7.9e+08	1.5e+06/ 8.4e+05	1.2e+02 /9.2e+02	6.4e+04/ 1.7e+04	1.2e+02 /2.9e+02	2.3e+06/ 2.0e+05
hashJoinPressure	F	1.6e+08 /4.3e+08	1.8e+08 /7.7e+08	1.7e+06/ 6.5e+05	1.3e+02 /7.2e+02	1.0e+05/ 1.6e+04	1.2e+02 /1.7e+02	2.2e+06/ 1.1e+05
scaleCosts	M	2.0e+08 /1.3e+09	3.5e+08 /2.4e+09	5.1e+05 /2.3e+07	7.3e+04 /2.9e+05	1.7e+04 /5.1e+04	2.6e+03 /4.6e+04	2.3e+06/ 1.2e+06
sumCost	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.7e+06/ 1.2e+06	1.4e+02 /8.3e+02	7.2e+04/ 2.2e+04	1.4e+02 /4.6e+02	2.3e+06/ 3.3e+05
sumMemory	F	1.4e+08 /4.5e+08	1.8e+08 /7.8e+08	1.5e+06/ 1.1e+06	2.4e+02 /9.0e+02	6.8e+04/ 2.1e+04	2.4e+02 /4.3e+02	2.3e+06/ 3.0e+05
sumRows	F	1.5e+08 /4.4e+08	2.3e+08 /8.3e+08	1.6e+06/ 1.2e+06	1.5e+02 /7.6e+02	5.6e+04/ 2.2e+04	1.5e+02 /3.8e+02	2.2e+06/ 3.3e+05
Total		1.3e+09/5.3e+09	1.8e+09/9.5e+09	1.1e+07/4.8e+07	1.7e+05/9.6e+05	5.1e+05/2.1e+05	5.7e+03/8.8e+04	1.8e+07/3.5e+06

Table 12: Per-pass performance for DecisionTree, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
classify Batch	F	5/7	29%	3.122 $\pm$ 0.0052	4.229 $\pm$ 0.0049	2.611$\pm$0.0056	20.84 $\pm$ 0.010	1.20$\times$	1.35$\times$	1.62$\times$	0.20 $\times$
classify Depth	F	4/7	43%	0.0447 $\pm$ 0.00001	0.0606 $\pm$ 0.00002	0.0391$\pm$0.00123	0.2999 $\pm$ 0.00076	1.14$\times$	1.35$\times$	1.55$\times$	0.20 $\times$
classify tree	F	5/7	29%	0.0447 $\pm$ 0.00001	0.0606 $\pm$ 0.00002	0.0391$\pm$0.00005	0.2975 $\pm$ 0.00119	1.14$\times$	1.36$\times$	1.55$\times$	0.20 $\times$
countFeatureUses	F	3/7	57%	0.0477 $\pm$ 0.00003	0.1405 $\pm$ 0.00019	0.0316$\pm$0.00002	0.3784 $\pm$ 0.00091	1.51$\times$	2.94$\times$	4.44$\times$	0.37 $\times$
countLeaves	F	2/7	71%	0.0455 $\pm$ 0.00080	0.1382 $\pm$ 0.00006	0.0249$\pm$0.00001	0.3693 $\pm$ 0.00118	1.82$\times$	3.04$\times$	5.55$\times$	0.37 $\times$
countNodes	F	2/7	71%	0.0452 $\pm$ 0.00002	0.1385 $\pm$ 0.00129	0.0260$\pm$0.00003	0.3688 $\pm$ 0.00165	1.74$\times$	3.06$\times$	5.32$\times$	0.38 $\times$
countSmallLeaves	F	3/7	57%	0.0485 $\pm$ 0.00003	0.1346 $\pm$ 0.00006	0.0330$\pm$0.00002	0.3716 $\pm$ 0.00119	1.47$\times$	2.78$\times$	4.08$\times$	0.36 $\times$
inferenceCost	F	2/7	71%	0.0484 $\pm$ 0.00083	0.1370 $\pm$ 0.00144	0.0290$\pm$0.00001	0.3701 $\pm$ 0.00231	1.67$\times$	2.83$\times$	4.73$\times$	0.37 $\times$
sumImpurity	F	3/7	57%	0.0477 $\pm$ 0.00067	0.1374 $\pm$ 0.00019	0.0322$\pm$0.00013	0.3810 $\pm$ 0.00175	1.48$\times$	2.88$\times$	4.27$\times$	0.36 $\times$
sumPathLengths	F	3/7	57%	0.0505 $\pm$ 0.00002	0.1370 $\pm$ 0.00145	0.0335$\pm$0.00001	0.3875 $\pm$ 0.00330	1.51$\times$	2.71$\times$	4.09$\times$	0.35 $\times$
sumSamples	F	3/7	57%	0.0458 $\pm$ 0.00001	0.1331 $\pm$ 0.00003	0.0317$\pm$0.00002	0.3839 $\pm$ 0.00118	1.45$\times$	2.90$\times$	4.20$\times$	0.35 $\times$
treeDepth	F	2/7	71%	0.0488 $\pm$ 0.00001	0.1368 $\pm$ 0.00121	0.0280$\pm$0.00005	0.3700 $\pm$ 0.00120	1.74$\times$	2.80$\times$	4.89$\times$	0.37 $\times$
Total				3.640	5.583	2.959	24.81	1.23$\times$	1.53$\times$	1.89$\times$	0.22 $\times$
Geomean								1.47$\times$	2.39$\times$	3.51$\times$	0.31 $\times$

Table 13: Per-pass PAPI counters for DecisionTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
classify Batch	F	1.6e+10/ 1.3e+10	3.3e+10 /5.9e+10	4.6e+08/ 2.7e+07	9.9e+04/ 4.1e+04	1.4e+06 /2.0e+06	9.9e+04/ 3.9e+04	9.3e+08/ 1.7e+07
classify Depth	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.5e+06/ 4.1e+05	8.0e+02/ 3.7e+02	2.1e+04 /2.8e+04	8.0e+02/ 3.5e+02	1.3e+07/ 2.7e+05
classify tree	F	2.3e+08/ 2.0e+08	4.7e+08 /8.5e+08	6.5e+06/ 4.1e+05	7.8e+02/ 4.5e+02	1.9e+04 /2.8e+04	7.8e+02/ 4.3e+02	1.3e+07/ 2.8e+05
countFeatureUses	F	2.4e+08/ 1.6e+08	7.5e+08 /7.8e+08	7.4e+06/ 3.4e+05	8.5e+02/ 3.5e+02	3.6e+04/ 1.4e+04	8.4e+02/ 3.6e+02	1.5e+07/ 2.0e+06
countLeaves	F	2.3e+08/ 1.3e+08	6.0e+08 /6.1e+08	7.4e+06/ 1.0e+05	8.6e+02/ 3.0e+02	4.4e+04/ 7.9e+03	8.5e+02/ 3.0e+02	1.4e+07/ 3.4e+05
countNodes	F	2.3e+08/ 1.3e+08	6.1e+08 /6.2e+08	7.2e+06/ 1.1e+05	7.4e+02/ 3.1e+02	5.3e+04/ 7.9e+03	7.4e+02/ 3.1e+02	1.4e+07/ 3.5e+05
countSmallLeaves	F	2.5e+08/ 1.7e+08	7.1e+08 /8.4e+08	5.9e+06/ 8.4e+04	9.9e+02/ 5.8e+02	3.3e+04/ 1.5e+04	9.9e+02/ 5.9e+02	1.5e+07/ 1.9e+06
inferenceCost	F	2.5e+08/ 1.5e+08	6.1e+08/ 6.1e+08	8.1e+06/ 1.2e+05	8.8e+02/ 4.4e+02	1.7e+04/ 7.8e+03	8.8e+02/ 4.3e+02	1.4e+07/ 3.3e+05
sumImpurity	F	2.4e+08/ 1.6e+08	7.6e+08 /7.8e+08	7.6e+06/ 1.1e+06	1.1e+03/ 7.8e+02	3.6e+04/ 1.3e+04	1.1e+03/ 7.8e+02	1.4e+07/ 2.1e+06
sumPathLengths	F	2.6e+08/ 1.7e+08	8.0e+08 /9.0e+08	7.4e+06/ 4.7e+05	1.6e+03/ 8.5e+02	4.0e+04/ 1.4e+04	1.5e+03/ 8.5e+02	1.5e+07/ 2.0e+06
sumSamples	F	2.3e+08/ 1.6e+08	6.4e+08 /7.7e+08	6.2e+06/ 2.4e+05	9.6e+02/ 4.2e+02	4.1e+04/ 1.4e+04	9.6e+02/ 4.3e+02	1.5e+07/ 2.0e+06
treeDepth	F	2.5e+08/ 1.4e+08	6.1e+08/ 6.1e+08	8.2e+06/ 1.1e+05	1.0e+03/ 4.2e+02	1.7e+04/ 7.9e+03	1.0e+03/ 4.0e+02	1.4e+07/ 3.4e+05
Total		1.8e+10/1.5e+10	4.0e+10/6.8e+10	5.4e+08/3.0e+07	1.1e+05/4.7e+04	1.8e+06/2.1e+06	1.1e+05/4.4e+04	1.1e+09/2.9e+07

Table 14: Per-pass PAPI counters for DecisionTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
classify Batch	F	2.1e+10 /1.1e+11	5.1e+10 /1.9e+11	1.7e+07 /4.7e+07	2.1e+04 /1.3e+05	3.4e+05 /2.2e+06	2.1e+04/ 1.9e+04	8.6e+08/ 1.8e+07
classify Depth	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.4e+05	2.5e+02 /3.2e+03	4.7e+03 /3.1e+04	2.5e+02 /7.3e+02	1.2e+07/ 3.0e+05
classify tree	F	3.1e+08 /1.5e+09	7.4e+08 /2.7e+09	2.5e+05 /6.8e+05	2.1e+02 /1.7e+03	4.8e+03 /3.2e+04	2.1e+02/ 1.8e+02	1.2e+07/ 2.9e+05
countFeatureUses	F	7.1e+08 /1.9e+09	1.3e+09 /3.1e+09	2.0e+06 /3.2e+06	8.0e+02 /2.0e+03	5.8e+03 /6.9e+04	8.0e+02/ 7.3e+02	9.3e+06/ 1.2e+06
countLeaves	F	7.0e+08 /1.9e+09	1.2e+09 /2.9e+09	4.4e+06/ 6.7e+05	3.4e+03/ 2.6e+03	1.0e+04 /3.2e+04	3.4e+03/ 1.9e+02	9.2e+06/ 3.4e+05
countNodes	F	7.0e+08 /1.9e+09	1.2e+09 /3.0e+09	4.4e+06/ 6.8e+05	7.9e+02 /1.8e+03	5.3e+03 /3.2e+04	7.9e+02/ 2.6e+02	9.2e+06/ 3.4e+05
countSmallLeaves	F	6.8e+08 /1.9e+09	1.3e+09 /2.9e+09	1.1e+06 /3.4e+06	1.0e+03 /1.8e+03	4.6e+03 /7.3e+04	1.0e+03/ 6.1e+02	9.3e+06/ 1.2e+06
inferenceCost	F	6.9e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	5.2e+02 /1.5e+03	4.5e+03 /3.1e+04	5.3e+02/ 3.1e+02	9.2e+06/ 3.6e+05
sumImpurity	F	7.0e+08 /1.9e+09	1.3e+09 /3.0e+09	2.2e+06 /3.5e+06	6.0e+02/ 3.5e+02	6.1e+03 /7.5e+04	6.0e+02/ 3.2e+02	9.4e+06/ 1.2e+06
sumPathLengths	F	7.0e+08 /2.0e+09	1.3e+09 /3.1e+09	1.5e+06 /3.6e+06	4.6e+02 /3.7e+03	4.1e+03 /7.9e+04	4.6e+02 /1.2e+03	9.2e+06/ 1.2e+06
sumSamples	F	6.7e+08 /2.0e+09	1.2e+09 /3.0e+09	1.1e+06 /3.6e+06	7.2e+02 /2.5e+03	4.1e+03 /7.7e+04	7.1e+02 /8.8e+02	9.3e+06/ 1.2e+06
treeDepth	F	6.9e+08 /1.9e+09	1.3e+09 /3.0e+09	4.4e+06/ 6.2e+05	2.0e+02 /1.5e+03	4.6e+03 /3.1e+04	2.0e+02 /2.4e+02	9.2e+06/ 3.7e+05
Total		2.8e+10/1.3e+11	6.4e+10/2.2e+11	4.3e+07/6.9e+07	3.0e+04/1.5e+05	4.0e+05/2.8e+06	3.0e+04/2.5e+04	9.6e+08/2.6e+07

Table 15: Per-pass performance for DomTree, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1 $\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
SumArea	F	6/15	60%	0.0346 $\pm$ 0.00002	0.0680 $\pm$ 0.00033	0.0191$\pm$0.00003	0.0903 $\pm$ 0.00123	1.82$\times$	1.96$\times$	3.57$\times$	0.75 $\times$
computeWidths	M	14/15	7%	0.0297$\pm$0.00004	0.0311 $\pm$ 0.00004	0.0444 $\pm$ 0.00003	0.0883 $\pm$ 0.00006	0.67 $\times$	1.05 $\times$	0.70 $\times$	0.35 $\times$
count styled	F	3/15	80%	0.0335 $\pm$ 0.00001	0.0678 $\pm$ 0.00013	0.0116$\pm$0.00001	0.0879 $\pm$ 0.00130	2.89$\times$	2.02$\times$	5.86$\times$	0.77 $\times$
find max Bottom	F	5/15	67%	0.0335 $\pm$ 0.00068	0.0674 $\pm$ 0.00001	0.0183$\pm$0.00082	0.0880 $\pm$ 0.00097	1.83$\times$	2.01$\times$	3.68$\times$	0.77 $\times$
scaleLayout	M	15/15	0%	0.0283$\pm$0.00015	0.0307 $\pm$ 0.00005	0.0346 $\pm$ 0.00004	0.0849 $\pm$ 0.00007	0.82 $\times$	1.09 $\times$	0.89 $\times$	0.36 $\times$
sumTextWidth	F	3/15	80%	0.0333 $\pm$ 0.00001	0.0684 $\pm$ 0.00001	0.0116$\pm$0.00002	0.0882 $\pm$ 0.00014	2.87$\times$	2.05$\times$	5.90$\times$	0.78 $\times$
Total				0.1929	0.3336	0.1395	0.5277	1.38$\times$	1.73$\times$	2.39$\times$	0.63 $\times$
Geomean								1.57$\times$	1.63$\times$	2.56$\times$	0.59 $\times$

Table 16: Per-pass PAPI counters for DomTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
SumArea	F	1.8e+08/ 9.7e+07	2.9e+08 /4.9e+08	2.7e+06/ 7.2e+05	3.3e+02/ 1.1e+02	1.7e+05/ 1.2e+04	3.3e+02/ 1.1e+02	1.4e+07/ 1.8e+06
computeWidths	M	6.5e+07 /1.4e+08	1.4e+08 /4.6e+08	1.2e+06 /1.5e+06	5.0e+04 /8.1e+04	4.9e+03 /9.9e+03	1.5e+03 /1.7e+04	1.2e+06/ 4.3e+05
count styled	F	1.7e+08/ 5.9e+07	2.6e+08 /2.6e+08	2.1e+06/ 2.8e+04	8.2e+02/ 7.4e+01	2.9e+05/ 4.9e+03	8.2e+02/ 7.3e+01	1.4e+07/ 6.4e+05
find max Bottom	F	1.7e+08/ 9.3e+07	2.5e+08 /4.1e+08	2.4e+06/ 8.4e+05	4.5e+02/ 7.2e+01	1.6e+05/ 9.3e+03	4.5e+02/ 7.1e+01	1.4e+07/ 1.4e+06
scaleLayout	M	5.9e+07 /8.8e+07	8.8e+07 /2.9e+08	1.1e+06/ 7.3e+05	7.5e+04/ 4.8e+04	1.6e+04/ 8.5e+03	3.2e+03 /1.5e+04	1.3e+06/ 4.5e+05
sumTextWidth	F	1.7e+08/ 5.9e+07	2.0e+08 /2.4e+08	2.4e+06/ 1.9e+04	6.3e+02/ 7.6e+01	4.3e+05/ 4.2e+03	6.3e+02/ 7.6e+01	1.4e+07/ 6.5e+05
Total		8.1e+08/5.3e+08	1.2e+09/2.2e+09	1.2e+07/3.9e+06	1.3e+05/1.3e+05	1.1e+06/4.9e+04	6.9e+03/3.2e+04	5.7e+07/5.4e+06

Table 17: Per-pass PAPI counters for DomTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
SumArea	F	3.5e+08 /4.6e+08	4.4e+08 /1.8e+09	8.2e+06/ 2.2e+06	1.9e+02 /9.1e+02	1.2e+04 /3.5e+04	1.9e+02 /9.2e+02	1.0e+07/ 1.3e+06
computeWidths	M	6.8e+07 /3.6e+08	1.5e+08 /9.0e+08	1.4e+06 /3.3e+06	5.9e+04 /2.4e+05	4.3e+03 /1.5e+04	6.0e+02 /2.3e+04	1.1e+06/ 4.4e+05
count styled	F	3.4e+08 /4.5e+08	4.3e+08 /1.7e+09	1.0e+07/ 1.2e+06	1.8e+02 /3.3e+02	3.5e+04 /3.7e+04	1.8e+02 /2.8e+02	1.0e+07/ 4.1e+05
find max Bottom	F	3.4e+08 /4.5e+08	4.6e+08 /1.8e+09	1.0e+07/ 1.5e+06	6.6e+01 /7.4e+02	1.4e+04 /2.8e+04	7.1e+01 /7.5e+02	1.0e+07/ 9.6e+05
scaleLayout	M	6.7e+07 /3.4e+08	1.1e+08 /7.6e+08	9.7e+05 /2.9e+06	6.3e+04 /7.7e+04	3.8e+03 /1.2e+04	5.7e+02 /2.1e+04	1.2e+06/ 4.6e+05
sumTextWidth	F	3.5e+08 /4.5e+08	3.9e+08 /1.7e+09	1.1e+07/ 1.3e+06	3.7e+02 /7.9e+02	8.7e+04/ 4.0e+04	3.7e+02 /7.1e+02	1.0e+07/ 4.1e+05
Total		1.5e+09/2.5e+09	2.0e+09/8.6e+09	4.1e+07/1.2e+07	1.2e+05/3.2e+05	1.6e+05/1.7e+05	2.0e+03/4.7e+04	4.3e+07/4.0e+06

Table 18: Per-pass performance for KDTree, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
Find nearest Neighbour	F	13/17	24%	0.0270 $\pm$ 0.00076	0.0389 $\pm$ 0.00009	0.0211$\pm$0.00089	0.1174 $\pm$ 0.00175	1.28$\times$	1.44$\times$	1.85$\times$	0.33 $\times$
countInRange tight_box	F	13/17	24%	0.0271 $\pm$ 0.00001	0.0396 $\pm$ 0.00001	0.0196$\pm$0.00001	0.1230 $\pm$ 0.00143	1.38$\times$	1.46$\times$	2.02$\times$	0.32 $\times$
photonMappingBenchmark	F	12/17	29%	0.0416 $\pm$ 0.00001	0.0471 $\pm$ 0.00106	0.0410$\pm$0.00001	0.1317 $\pm$ 0.00176	1.02 $\times$	1.13$\times$	1.15$\times$	0.36 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0242 $\pm$ 0.00001	0.0380 $\pm$ 0.00001	0.0163$\pm$0.00000	0.1131 $\pm$ 0.00158	1.48$\times$	1.57$\times$	2.33$\times$	0.34 $\times$
sumMassInRange	F	14/17	18%	0.0264 $\pm$ 0.00001	0.0397 $\pm$ 0.00001	0.0200$\pm$0.00000	0.1231 $\pm$ 0.00001	1.32$\times$	1.50$\times$	1.98$\times$	0.32 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0280 $\pm$ 0.00002	0.0387 $\pm$ 0.00001	0.0206$\pm$0.00001	0.1129 $\pm$ 0.00020	1.36$\times$	1.38$\times$	1.88$\times$	0.34 $\times$
Total				0.1742	0.2419	0.1386	0.7211	1.26$\times$	1.39$\times$	1.74$\times$	0.34 $\times$
Geomean								1.30$\times$	1.41$\times$	1.83$\times$	0.33 $\times$

Table 19: Per-pass PAPI counters for KDTree (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
Find nearest Neighbour	F	1.4e+08/ 1.1e+08	4.7e+08 /6.3e+08	2.8e+06/ 1.6e+06	1.5e+02 /1.6e+02	4.9e+03 /1.6e+04	1.5e+02 /1.6e+02	7.8e+06/ 1.6e+06
countInRange tight_box	F	1.4e+08/ 9.9e+07	4.0e+08 /5.5e+08	2.0e+06/ 1.9e+06	1.1e+02/ 9.7e+01	2.9e+03 /1.6e+04	1.2e+02/ 9.7e+01	7.7e+06/ 1.6e+06
photonMappingBenchmark	F	2.1e+08/ 2.1e+08	1.0e+09 /1.2e+09	2.2e+06 /2.5e+06	2.4e+02 /4.5e+02	3.5e+03 /1.4e+04	2.4e+02 /4.5e+02	5.6e+06/ 1.4e+06
pointCloudNeighborhood	F	1.2e+08/ 8.3e+07	3.3e+08 /4.8e+08	2.5e+06/ 1.5e+06	6.8e+01 /9.6e+01	4.2e+03 /1.5e+04	7.1e+01 /9.5e+01	7.8e+06/ 1.4e+06
sumMassInRange	F	1.3e+08/ 1.0e+08	4.0e+08 /5.7e+08	2.4e+06 /3.0e+06	6.6e+01 /1.3e+02	3.8e+03 /1.9e+04	6.9e+01 /1.3e+02	7.7e+06/ 1.8e+06
twoPointCorrelation bin_8_16	F	1.4e+08/ 1.0e+08	4.8e+08 /6.1e+08	2.2e+06/ 6.6e+05	1.2e+02/ 1.1e+02	3.6e+03 /1.2e+04	1.2e+02/ 1.1e+02	7.7e+06/ 1.3e+06
Total		8.8e+08/7.0e+08	3.1e+09/4.0e+09	1.4e+07/1.1e+07	7.6e+02/1.0e+03	2.3e+04/9.3e+04	7.7e+02/1.0e+03	4.4e+07/9.1e+06

Table 20: Per-pass PAPI counters for KDTree (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Find nearest Neighbour	F	2.0e+08 /6.0e+08	4.9e+08 /1.3e+09	4.7e+06 /5.2e+06	2.2e+02 /1.6e+03	5.0e+03 /2.1e+04	2.2e+02 /1.6e+03	5.8e+06/ 1.5e+06
countInRange tight_box	F	2.0e+08 /6.2e+08	4.5e+08 /1.3e+09	5.2e+06 /5.8e+06	1.1e+02 /2.2e+03	1.7e+04 /2.2e+04	1.1e+02 /2.2e+03	5.6e+06/ 1.6e+06
photonMappingBenchmark	F	2.4e+08 /6.7e+08	1.1e+09 /2.0e+09	2.9e+06 /5.9e+06	4.1e+02 /1.8e+03	2.9e+03 /2.4e+04	4.1e+02 /1.8e+03	5.1e+06/ 1.5e+06
pointCloudNeighborhood	F	1.9e+08 /5.7e+08	4.1e+08 /1.3e+09	5.1e+06/ 4.7e+06	1.8e+02 /1.1e+03	5.5e+03 /1.9e+04	1.8e+02 /1.1e+03	5.9e+06/ 1.4e+06
sumMassInRange	F	2.0e+08 /6.2e+08	4.7e+08 /1.3e+09	5.1e+06 /6.3e+06	1.5e+02 /1.5e+03	1.5e+04 /2.6e+04	1.6e+02 /1.5e+03	5.6e+06/ 1.6e+06
twoPointCorrelation bin_8_16	F	2.0e+08 /5.7e+08	6.1e+08 /1.4e+09	4.8e+06/ 3.7e+06	2.2e+02 /1.3e+03	9.4e+03 /1.8e+04	2.2e+02 /1.3e+03	5.8e+06/ 1.3e+06
Total		1.2e+09/3.7e+09	3.5e+09/8.4e+09	2.8e+07/3.2e+07	1.3e+03/9.6e+03	5.5e+04/1.3e+05	1.3e+03/9.4e+03	3.4e+07/8.8e+06

Table 21: Per-pass performance for LinearListReduction, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across -iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
reduction	F	2/11	82%	0.0290 $\pm$ 0.00002	0.1323 $\pm$ 0.00142	0.0050$\pm$0.00002	<i>OOM</i>	5.84$\times$	4.56$\times$	26.64$\times$	–
Total				0.0290	0.1323	0.0050	–	5.84$\times$	4.56$\times$	26.64$\times$	–
Geomean								5.84$\times$	4.56$\times$	26.64$\times$	–

Table 22: Per-pass PAPI counters for LinearListReduction (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
reduction	F	1.5e+08/ 2.5e+07	9.0e+07 /1.0e+08	1.5e+06/ 4.6e+04	3.5e+02/ 4.6e+01	2.8e+05/ 6.9e+03	3.5e+02/ 4.7e+01	1.2e+07/ 1.0e+06
Total		1.5e+08/2.5e+07	9.0e+07/1.0e+08	1.5e+06/4.6e+04	3.5e+02/4.6e+01	2.8e+05/6.9e+03	3.5e+02/4.7e+01	1.2e+07/1.0e+06

Table 23: Per-pass PAPI counters for LinearListReduction (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
reduction	F	6.7e+08/–	2.5e+08/–	4.0e+06/–	3.9e+02/–	3.8e+05/–	3.9e+02/–	2.7e+07/–
Total		6.7e+08/–	2.5e+08/–	4.0e+06/–	3.9e+02/–	3.8e+05/–	3.9e+02/–	2.7e+07/–

Table 24: Per-pass performance for `List`, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1 List	M	2/2	0%	0.2520±0.00119	<i>OOM</i>	0.3371±0.00215	<i>OOM</i>	0.75×	–	–	–
length List	M	1/2	50%	0.0408±0.00001	<i>OOM</i>	0.0214±0.00001	<i>OOM</i>	1.90×	–	–	–
sumList	F	2/2	0%	0.0440±0.00014	<i>OOM</i>	0.0450±0.00002	<i>OOM</i>	0.98×	–	–	–
sumList tail recursive	F	2/2	0%	0.0442±0.00002	<i>OOM</i>	0.0445±0.00001	<i>OOM</i>	0.99×	–	–	–
Total				0.3809	–	0.4480	–	0.85×	–	–	–
Geomean								1.08×	–	–	–

Table 25: Per-pass PAPI counters for `List` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1 List	M	5.9e+08 /9.6e+08	1.9e+09 /2.8e+09	8.0e+06/ 1.2e+06	2.9e+05 /5.7e+05	9.9e+03 /5.6e+04	1.1e+04 /1.3e+04	9.1e+06/ 7.7e+06
length List	M	2.1e+08/ 1.1e+08	8.0e+08/ 8.0e+08	1.2e+05/ 9.8e+04	1.1e+02/ 7.3e+01	2.6e+03 /1.4e+04	1.1e+02/ 7.3e+01	1.3e+07/ 1.1e+06
sumList	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	7.0e+05/ 5.4e+05	8.9e+01/ 7.0e+01	3.0e+03 /6.9e+04	9.2e+01/ 7.1e+01	1.3e+07/ 1.2e+07
sumList tail recursive	F	2.2e+08 /2.3e+08	9.0e+08 /1.0e+09	5.8e+05/ 5.6e+05	1.3e+02 /1.8e+02	3.0e+03 /5.9e+04	1.3e+02 /1.8e+02	1.3e+07/ 1.2e+07
Total		1.2e+09/1.5e+09	4.5e+09/5.6e+09	9.4e+06/2.4e+06	2.9e+05/5.7e+05	1.9e+04/2.0e+05	1.1e+04/1.4e+04	4.8e+07/3.3e+07

Table 26: Per-pass PAPI counters for `List` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
Total		–/–	–/–	–/–	–/–	–/–	–/–	–/–

Table 27: Per-pass performance for `MonoTree`, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add1Tree	M	3/3	0%	0.0453±0.00017	0.0603±0.00041	0.0427±0.00036	0.1125±0.00114	1.06×	1.33×	1.41×	0.54×
sumTree	F	3/3	0%	0.0088±0.00001	0.0406±0.00002	0.0114±0.00001	0.0805±0.00129	0.77×	4.62×	3.57×	0.50×
sumTree TailRec	F	3/3	0%	0.0082±0.00001	0.0402±0.00103	0.0111±0.00002	0.0806±0.00001	0.74×	4.89×	3.62×	0.50×
Total				0.0623	0.1411	0.0651	0.2737	0.96×	2.26×	2.17×	0.52×
Geomean								0.85×	3.11×	2.63×	0.51×

Table 28: Per-pass PAPI counters for `MonoTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add1Tree	M	1.6e+08/ 1.5e+08	5.0e+08 /5.9e+08	2.3e+05 /4.1e+05	4.0e+04 /4.5e+04	1.7e+03 /5.0e+03	9.9e+02 /1.6e+03	5.0e+05/ 4.5e+05
sumTree	F	4.5e+07 /5.8e+07	2.2e+08 /2.6e+08	8.2e+03 /3.5e+04	1.4e+02 /1.6e+02	5.9e+02 /3.9e+03	1.4e+02 /1.6e+02	7.7e+05/ 6.1e+05
sumTree TailRec	F	4.2e+07 /5.6e+07	1.9e+08 /2.2e+08	5.7e+03 /2.2e+04	1.7e+02 /1.8e+02	6.7e+02 /3.9e+03	1.7e+02 /1.8e+02	7.7e+05/ 6.0e+05
Total		2.5e+08/2.6e+08	9.1e+08/1.1e+09	2.4e+05/4.7e+05	4.0e+04/4.5e+04	2.9e+03/1.3e+04	1.3e+03/1.9e+03	2.0e+06/1.7e+06

Table 29: Per-pass PAPI counters for `MonoTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add1Tree	M	2.4e+08 /5.0e+08	7.1e+08 /1.2e+09	1.5e+05 /9.8e+05	3.0e+04 /6.0e+04	1.6e+03 /1.6e+04	1.3e+03 /3.1e+03	4.0e+05/ 3.9e+05
sumTree	F	2.1e+08 /4.1e+08	4.3e+08 /6.3e+08	9.5e+04 /8.9e+05	1.8e+02 /2.4e+02	1.4e+03 /1.3e+04	1.8e+02 /2.4e+02	3.4e+05 /3.6e+05
sumTree TailRec	F	2.0e+08 /4.1e+08	4.1e+08 /6.4e+08	7.0e+04 /8.7e+05	5.5e+02/ 1.1e+02	1.2e+03 /1.3e+04	5.4e+02/ 1.1e+02	3.2e+05 /3.5e+05
Total		6.5e+08/1.3e+09	1.6e+09/2.5e+09	3.2e+05/2.7e+06	3.1e+04/6.1e+04	4.2e+03/4.2e+04	2.0e+03/3.5e+03	1.1e+06/1.1e+06

Table 30: Per-pass performance for `ObjectGraph`, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
countLargeItems	F	3/5	40%	0.0123±0.00000	0.0445±0.00002	0.0103±0.00002	0.0463±0.00001	1.19×	3.63×	4.30×	0.96×
countMarked	F	3/5	40%	0.0122±0.00001	0.0444±0.00001	0.0107±0.00002	0.0467±0.00014	1.14×	3.63×	4.15×	0.95×
countSurvivors	F	4/5	20%	0.0133±0.00001	0.0446±0.00001	0.0128±0.00000	0.0468±0.00001	1.04×	3.36×	3.49×	0.95×
deadBytes	F	4/5	20%	0.0131±0.00000	0.0438±0.00015	0.0122±0.00000	0.0470±0.00002	1.07×	3.35×	3.58×	0.93×
liveBytes	F	4/5	20%	0.0129±0.00000	0.0432±0.00001	0.0123±0.00000	0.0470±0.00001	1.05×	3.36×	3.52×	0.92×
sumObjIds	F	3/5	40%	0.0122±0.00000	0.0432±0.00003	0.0103±0.00002	0.0471±0.00111	1.19×	3.54×	4.20×	0.92×
sweepUnmarked	M	5/5	0%	0.0719±0.00006	0.0887±0.00125	0.0815±0.00020	0.1865±0.00006	0.88×	1.23×	1.09×	0.48×
totalHeapSize	F	3/5	40%	0.0121±0.00055	0.0431±0.00049	0.0106±0.00010	0.0471±0.00004	1.14×	3.58×	4.07×	0.92×
touchHotObjects	M	5/5	0%	0.0708±0.00006	0.0883±0.00114	0.0805±0.00002	0.1919±0.00020	0.88×	1.25×	1.10×	0.46×
Total				0.2308	0.4840	0.2413	0.7065	0.96×	2.10×	2.01×	0.69×
Geomean								1.06×	2.77×	2.93×	0.80×

Table 31: Per-pass PAPI counters for `ObjectGraph` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
countLargeItems	F	6.2e+07/ 5.2e+07	2.6e+08 /2.8e+08	1.2e+06/ 2.9e+05	2.1e+02/ 1.6e+02	3.2e+03 /3.7e+03	2.1e+02/ 1.6e+02	2.8e+06/ 6.2e+05
countMarked	F	6.2e+07/ 5.4e+07	2.6e+08 /2.8e+08	1.2e+06/ 2.3e+05	1.6e+02 /1.6e+02	3.2e+03 /3.8e+03	1.6e+02 /1.6e+02	2.8e+06/ 6.2e+05
countSurvivors	F	6.7e+07/ 6.5e+07	3.1e+08 /3.8e+08	9.5e+05/ 1.4e+05	2.5e+02/ 1.6e+02	2.2e+03 /5.1e+03	2.5e+02/ 1.6e+02	2.7e+06/ 1.2e+06
deadBytes	F	6.6e+07/ 6.2e+07	2.7e+08 /3.5e+08	7.2e+05/ 1.1e+05	1.2e+02 /1.5e+02	4.3e+03 /5.3e+03	1.2e+02 /1.6e+02	2.7e+06/ 1.2e+06
liveBytes	F	6.5e+07/ 6.2e+07	2.7e+08 /3.5e+08	8.4e+05/ 9.7e+04	1.5e+02 /1.5e+02	2.3e+03 /5.2e+03	1.5e+02 /1.5e+02	2.7e+06/ 1.2e+06
sumObjIds	F	6.2e+07/ 5.2e+07	2.5e+08 /2.7e+08	1.1e+06/ 6.8e+04	1.4e+02/ 1.2e+02	2.7e+03 /4.3e+03	1.4e+02/ 1.2e+02	2.8e+06/ 6.1e+05
sweepUnmarked	M	1.9e+08 /2.4e+08	5.5e+08 /8.8e+08	1.0e+06/ 5.0e+05	9.4e+04 /1.1e+05	4.2e+03 /1.1e+04	3.1e+03 /7.0e+03	2.0e+06/ 1.0e+06
totalHeapSize	F	6.1e+07/ 5.4e+07	2.5e+08 /2.7e+08	1.2e+06/ 3.5e+05	1.7e+02/ 1.3e+02	2.4e+03 /3.6e+03	1.8e+02/ 1.3e+02	2.8e+06/ 6.1e+05
touchHotObjects	M	1.9e+08 /2.3e+08	5.7e+08 /9.0e+08	8.2e+05/ 4.8e+05	1.1e+05 /1.2e+05	4.8e+03 /1.1e+04	2.7e+03 /7.7e+03	2.0e+06/ 1.0e+06
Total		8.2e+08/8.7e+08	3.0e+09/4.0e+09	9.0e+06/2.3e+06	2.0e+05/2.2e+05	2.9e+04/5.3e+04	7.0e+03/1.6e+04	2.3e+07/8.0e+06

Table 32: Per-pass PAPI counters for `ObjectGraph` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
countLargeItems	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.6e+05	2.4e+02/ 6.4e+01	4.9e+03 /7.7e+03	2.4e+02/ 6.4e+01	1.8e+06/ 3.8e+05
countMarked	F	2.3e+08 /2.4e+08	4.6e+08 /7.9e+08	5.6e+05/ 4.8e+05	9.3e+01 /1.6e+02	5.0e+03 /9.5e+03	9.4e+01 /1.5e+02	1.8e+06/ 3.9e+05
countSurvivors	F	2.3e+08 /2.4e+08	5.5e+08 /8.2e+08	1.8e+05 /4.1e+05	1.6e+02 /1.9e+02	1.9e+03 /6.7e+03	1.6e+02 /1.9e+02	1.8e+06/ 6.7e+05
deadBytes	F	2.2e+08 /2.4e+08	4.9e+08 /8.1e+08	3.8e+05 /4.2e+05	1.2e+02 /2.8e+02	3.7e+03 /6.6e+03	1.3e+02 /2.8e+02	1.8e+06/ 6.9e+05
liveBytes	F	2.2e+08 /2.4e+08	5.0e+08 /8.1e+08	1.6e+05 /4.1e+05	9.1e+01 /2.8e+02	2.1e+03 /6.6e+03	9.2e+01 /2.8e+02	1.8e+06/ 7.0e+05
sumObjIds	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.5e+05/ 4.8e+05	2.3e+02/ 7.2e+01	4.7e+03 /8.7e+03	2.3e+02/ 7.2e+01	1.8e+06/ 3.6e+05
sweepUnmarked	M	2.7e+08 /7.7e+08	7.6e+08 /2.1e+09	2.1e+05 /2.5e+06	1.1e+05 /1.4e+05	3.5e+03 /2.3e+04	1.4e+03 /5.4e+03	1.8e+06/ 9.2e+05
totalHeapSize	F	2.2e+08 /2.4e+08	4.4e+08 /7.8e+08	5.4e+05/ 5.3e+05	8.9e+01 /3.4e+02	4.7e+03 /8.9e+03	8.9e+01 /3.4e+02	1.8e+06/ 3.7e+05
touchHotObjects	M	2.7e+08 /8.0e+08	7.7e+08 /2.2e+09	2.1e+05 /2.0e+06	1.4e+05/ 1.3e+05	3.5e+03 /2.1e+04	2.8e+03 /7.8e+03	1.8e+06/ 9.7e+05
Total		2.1e+09/3.2e+09	4.9e+09/9.9e+09	3.4e+06/7.7e+06	2.5e+05/2.7e+05	3.4e+04/9.9e+04	5.3e+03/1.5e+04	1.6e+07/5.5e+06

Table 33: Per-pass performance for `OctTree`, ADT has 16 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
barnesHutPotential	F	11/16	31%	0.0053 $\pm$ 0.00000	0.0074 $\pm$ 0.00000	0.0049$\pm$0.00000	<i>OOM</i>	1.08 $\times$	1.38$\times$	1.50$\times$	–
clearFlags	M	15/16	6%	0.0191 $\pm$ 0.00001	0.0186$\pm$0.00002	0.0219 $\pm$ 0.00002	<i>OOM</i>	0.87 $\times$	0.97 $\times$	0.85 $\times$	–
countActive	F	10/16	38%	0.0045 $\pm$ 0.00000	0.0058 $\pm$ 0.00000	0.0036$\pm$0.00000	<i>OOM</i>	1.26$\times$	1.29$\times$	1.62$\times$	–
countParticles	F	8/16	50%	0.0045 $\pm$ 0.00000	0.0056 $\pm$ 0.00000	0.0033$\pm$0.00000	<i>OOM</i>	1.35$\times$	1.26$\times$	1.70$\times$	–
fmmPotential	F	12/16	25%	0.0122 $\pm$ 0.00001	0.0122 $\pm$ 0.00000	0.0119$\pm$0.00000	<i>OOM</i>	1.02 $\times$	1.00 $\times$	1.03 $\times$	–
scaleEnergy	M	16/16	0%	0.0187$\pm$0.00012	0.0190 $\pm$ 0.00002	0.0221 $\pm$ 0.00142	<i>OOM</i>	0.85 $\times$	1.02 $\times$	0.86 $\times$	–
sumEnergy	F	12/16	25%	0.0046 $\pm$ 0.00000	0.0068 $\pm$ 0.00000	0.0040$\pm$0.00000	<i>OOM</i>	1.15$\times$	1.48$\times$	1.71$\times$	–
sumMass	F	10/16	38%	0.0044 $\pm$ 0.00009	0.0056 $\pm$ 0.00036	0.0033$\pm$0.00005	<i>OOM</i>	1.34$\times$	1.27$\times$	1.70$\times$	–
Total				0.0734	0.0811	0.0750	–	0.98 $\times$	1.11$\times$	1.08 $\times$	–
Geomean								1.10$\times$	1.20$\times$	1.32$\times$	–

Table 34: Per-pass PAPI counters for `OctTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
barnesHutPotential	F	2.7e+07/ 2.5e+07	1.0e+08 /1.3e+08	7.7e+03 /8.7e+04	6.4e+01/ 4.4e+01	2.8e+02 /3.1e+03	6.5e+01/ 4.7e+01	5.3e+05/ 1.7e+05
clearFlags	M	4.6e+07 /5.5e+07	1.0e+08 /2.2e+08	1.1e+05 /3.3e+05	2.0e+04 /4.7e+04	1.1e+03 /5.2e+03	1.1e+03 /1.6e+03	5.6e+05/ 2.7e+05
countActive	F	2.3e+07/ 1.8e+07	7.3e+07/ 7.2e+07	1.0e+05/ 5.2e+04	8.1e+01/ 2.1e+01	5.2e+02 /1.8e+03	8.1e+01/ 2.3e+01	5.5e+05/ 1.1e+03
countParticles	F	2.3e+07/ 1.7e+07	7.1e+07/ 7.1e+07	1.1e+05/ 2.5e+04	6.5e+01/ 1.3e+01	5.1e+02 /8.3e+02	6.8e+01/ 1.3e+01	5.4e+05/ 1.2e+02
fmmPotential	F	6.2e+07/ 6.1e+07	2.0e+08 /2.3e+08	3.2e+03 /1.1e+05	9.4e+01 /1.2e+02	3.0e+02 /3.7e+03	9.6e+01 /1.2e+02	4.2e+05/ 2.0e+05
scaleEnergy	M	4.4e+07 /5.5e+07	1.1e+08 /2.3e+08	1.7e+05 /3.7e+05	3.1e+04 /6.4e+04	1.1e+03 /6.0e+03	1.0e+03 /1.9e+03	5.6e+05/ 2.8e+05
sumEnergy	F	2.3e+07/ 2.0e+07	9.6e+07 /1.1e+08	4.2e+04 /8.6e+04	5.0e+01 /5.2e+01	3.1e+02 /3.1e+03	5.4e+01 /5.5e+01	5.5e+05/ 1.9e+05
sumMass	F	2.2e+07/ 1.7e+07	7.1e+07 /7.9e+07	8.7e+04/ 3.5e+04	5.8e+01/ 3.5e+01	6.9e+02 /1.3e+03	6.0e+01/ 3.6e+01	5.7e+05/ 2.5e+04
Total		2.7e+08/2.7e+08	8.2e+08/1.1e+09	6.4e+05/1.1e+06	5.2e+04/1.1e+05	4.8e+03/2.5e+04	2.5e+03/3.8e+03	4.3e+06/1.1e+06

Table 35: Per-pass PAPI counters for `OctTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
barnesHutPotential	F	3.7e+07/–	1.1e+08/–	7.9e+03/–	6.2e+01/–	2.8e+02/–	6.2e+01/–	4.9e+05/–
clearFlags	M	4.3e+07/–	1.0e+08/–	9.7e+04/–	2.6e+04/–	9.5e+02/–	9.3e+02/–	5.7e+05/–
countActive	F	2.9e+07/–	8.8e+07/–	3.5e+04/–	8.6e+01/–	3.0e+02/–	8.8e+01/–	5.1e+05/–
countParticles	F	2.8e+07/–	8.1e+07/–	5.1e+04/–	3.9e+01/–	3.5e+02/–	3.9e+01/–	5.3e+05/–
fmmPotential	F	6.2e+07/–	2.1e+08/–	5.3e+03/–	6.2e+01/–	2.7e+02/–	6.4e+01/–	4.4e+05/–
scaleEnergy	M	4.5e+07/–	1.2e+08/–	1.2e+05/–	2.6e+04/–	1.0e+03/–	9.1e+02/–	5.6e+05/–
sumEnergy	F	3.5e+07/–	1.1e+08/–	9.1e+03/–	7.2e+01/–	3.1e+02/–	7.6e+01/–	5.1e+05/–
sumMass	F	2.8e+07/–	8.3e+07/–	6.4e+04/–	6.6e+01/–	3.4e+02/–	7.0e+01/–	5.5e+05/–
Total		3.1e+08/–	9.1e+08/–	3.9e+05/–	5.2e+04/–	3.8e+03/–	2.2e+03/–	4.2e+06/–

Table 36: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
addConstPW	M	8/8	0%	0.1184$\pm$0.00014	0.1320 $\pm$ 0.00117	0.1291 $\pm$ 0.00129	0.3974 $\pm$ 0.00167	0.92 $\times$	1.11$\times$	1.02 $\times$	0.33 $\times$
autorefineMaxLevel	F	4/8	50%	0.0199 $\pm$ 0.00066	0.0444 $\pm$ 0.00113	0.0195$\pm$0.00004	0.2380 $\pm$ 0.00173	1.02 $\times$	2.23$\times$	2.28$\times$	0.19 $\times$
compressMass	F	3/8	62%	0.0190 $\pm$ 0.00001	0.0449 $\pm$ 0.00001	0.0115$\pm$0.00001	0.2379 $\pm$ 0.00003	1.65$\times$	2.37$\times$	3.90$\times$	0.19 $\times$
diffPW	M	8/8	0%	0.1203$\pm$0.00109	0.1329 $\pm$ 0.00123	0.1286 $\pm$ 0.00026	0.3988 $\pm$ 0.00163	0.94 $\times$	1.10$\times$	1.03 $\times$	0.33 $\times$
lbDeuxLoadProxy	F	5/8	38%	0.0209 $\pm$ 0.00068	0.0466 $\pm$ 0.00003	0.0184$\pm$0.00001	0.2414 $\pm$ 0.00176	1.13$\times$	2.23$\times$	2.53$\times$	0.19 $\times$
norm2Estimate	F	5/8	38%	0.0243 $\pm$ 0.00001	0.0543 $\pm$ 0.00039	0.0176$\pm$0.00005	0.2000 $\pm$ 0.00230	1.38$\times$	2.23$\times$	3.09$\times$	0.27 $\times$
pmapCutHistogram	F	4/8	50%	0.0314 $\pm$ 0.00001	0.0551 $\pm$ 0.00126	0.0125$\pm$0.00000	0.2455 $\pm$ 0.00167	2.52$\times$	1.76$\times$	4.43$\times$	0.22 $\times$
truncateTolViolations	F	3/8	62%	0.0192 $\pm$ 0.00001	0.0456 $\pm$ 0.00001	0.0115$\pm$0.00001	0.2048 $\pm$ 0.00111	1.67$\times$	2.37$\times$	3.97$\times$	0.22 $\times$
Total				0.3734	0.5558	0.3487	2.164	1.07 $\times$	1.49$\times$	1.59$\times$	0.26 $\times$
Geomean								1.32$\times$	1.85$\times$	2.45$\times$	0.24 $\times$

Table 37: Per-pass PAPI counters for `PiecewiseFunctions` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
addConstPW	M	2.7e+08 /3.2e+08	5.7e+08 /1.2e+09	1.3e+06 /2.1e+06	1.7e+05 /2.7e+05	1.1e+04 /2.5e+04	1.4e+04 /2.5e+04	4.4e+06/ 1.8e+06
autorefineMaxLevel	F	1.0e+08/ 9.9e+07	2.5e+08 /3.7e+08	2.5e+06/ 2.0e+05	4.0e+02/ 5.0e+01	6.6e+03/ 6.2e+03	4.1e+02/ 5.0e+01	6.1e+06/ 9.0e+05
compressMass	F	9.6e+07/ 5.8e+07	2.5e+08 /2.8e+08	2.1e+06/ 2.9e+04	5.8e+02/ 2.5e+02	1.9e+04/ 4.8e+03	5.8e+02/ 2.6e+02	6.3e+06/ 6.0e+05
diffPW	M	2.8e+08 /3.2e+08	5.9e+08 /1.3e+09	1.1e+06 /1.9e+06	1.5e+05 /2.2e+05	1.1e+04 /2.3e+04	1.0e+04 /2.9e+04	4.4e+06/ 1.8e+06
lbDeuxLoadProxy	F	1.1e+08/ 9.4e+07	3.3e+08 /4.9e+08	2.4e+06/ 3.1e+05	7.1e+02/ 1.2e+02	1.5e+04/ 8.8e+03	7.1e+02/ 1.2e+02	6.3e+06/ 1.5e+06
norm2Estimate	F	1.2e+08/ 8.9e+07	3.4e+08 /4.4e+08	1.5e+06/ 4.3e+05	2.7e+02/ 1.4e+02	1.1e+04/ 9.4e+03	2.7e+02/ 1.4e+02	6.2e+06/ 1.5e+06
pmapCutHistogram	F	1.6e+08/ 6.3e+07	2.8e+08 /3.7e+08	1.7e+06/ 6.1e+04	1.3e+03/ 2.6e+02	1.0e+05/ 7.4e+03	1.3e+03/ 2.6e+02	4.8e+06/ 1.2e+06
truncateTolViolations	F	9.8e+07/ 5.8e+07	2.5e+08 /2.9e+08	2.1e+06/ 3.0e+04	4.6e+02/ 1.7e+02	1.7e+04/ 4.8e+03	4.6e+02/ 1.7e+02	6.3e+06/ 6.2e+05
Total		1.2e+09/1.1e+09	2.8e+09/4.7e+09	1.5e+07/5.1e+06	3.1e+05/4.9e+05	2.0e+05/9.0e+04	2.8e+04/5.5e+04	4.5e+07/1.0e+07

Table 38: Per-pass PAPI counters for `PiecewiseFunctions` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
addConstPW	M	3.3e+08 /1.7e+09	7.8e+08 /3.3e+09	3.4e+05 /4.8e+06	1.9e+05 /4.0e+05	5.7e+03 /3.9e+04	5.1e+03 /2.9e+04	4.2e+06/ 1.9e+06
autorefineMaxLevel	F	2.3e+08 /1.2e+09	4.9e+08 /1.3e+09	2.9e+05 /2.1e+06	2.4e+02 /1.3e+03	1.8e+03 /4.1e+04	2.4e+02 /4.1e+02	4.1e+06/ 6.7e+05
compressMass	F	2.3e+08 /1.2e+09	4.7e+08 /1.2e+09	3.0e+05 /1.3e+06	2.3e+02 /1.5e+03	1.7e+03 /3.0e+04	2.3e+02/ 2.1e+02	4.1e+06/ 3.7e+05
diffPW	M	3.4e+08 /1.7e+09	8.1e+08 /3.3e+09	3.5e+05 /5.1e+06	1.7e+05 /3.2e+05	6.5e+03 /3.6e+04	2.4e+03 /2.8e+04	4.2e+06/ 1.8e+06
lbDeuxLoadProxy	F	2.4e+08 /1.2e+09	5.3e+08 /1.3e+09	1.6e+05 /2.6e+06	1.2e+02 /2.8e+02	1.6e+03 /4.2e+04	1.2e+02 /2.7e+02	4.1e+06/ 1.0e+06
norm2Estimate	F	2.8e+08 /1.0e+09	5.1e+08 /1.3e+09	2.5e+05 /2.4e+06	1.4e+02 /7.7e+02	1.7e+03 /3.9e+04	1.4e+02 /6.9e+02	3.9e+06/ 9.7e+05
pmapCutHistogram	F	2.8e+08 /1.2e+09	5.0e+08 /1.3e+09	8.1e+05 /2.0e+06	7.9e+02 /2.1e+03	1.1e+04 /3.7e+04	7.9e+02/ 4.4e+02	3.9e+06/ 6.7e+05
truncateTolViolations	F	2.3e+08 /1.0e+09	4.4e+08 /1.3e+09	3.2e+05 /1.3e+06	2.5e+02 /3.4e+03	3.8e+03 /2.7e+04	2.5e+02 /9.4e+02	4.0e+06/ 3.7e+05
Total		2.1e+09/1.0e+10	4.5e+09/1.4e+10	2.8e+06/2.2e+07	3.6e+05/7.3e+05	3.3e+04/2.9e+05	9.3e+03/6.0e+04	3.3e+07/7.8e+06

Table 39: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
add 1 tree	M	5/5	0%	0.0831 $\pm$ 0.00006	0.0934 $\pm$ 0.00020	0.0778$\pm$0.00101	0.3430 $\pm$ 0.00245	1.07 $\times$	1.12$\times$	1.20$\times$	0.27 $\times$
sum tree	F	5/5	0%	0.0278 $\pm$ 0.00024	0.0503 $\pm$ 0.00002	0.0262$\pm$0.00008	0.2729 $\pm$ 0.00005	1.06 $\times$	1.81$\times$	1.92$\times$	0.18 $\times$
Total				0.1109	0.1437	0.1040	0.6159	1.07 $\times$	1.30$\times$	1.38$\times$	0.23 $\times$
Geomean								1.06 $\times$	1.43$\times$	1.52$\times$	0.22 $\times$

Table 40: Per-pass PAPI counters for `TernaryTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
add 1 tree	M	2.7e+08/ 2.3e+08	7.5e+08 /9.8e+08	4.4e+05 /1.3e+06	6.8e+04 /1.4e+05	3.6e+03 /1.1e+04	2.8e+03/ 2.3e+03	1.5e+06/ 1.1e+06
sum tree	F	1.4e+08/ 1.3e+08	4.6e+08 /5.4e+08	8.2e+04 /3.6e+05	6.3e+01 /7.1e+01	7.9e+02 /9.0e+03	6.4e+01 /7.1e+01	1.8e+06/ 1.3e+06
Total		4.1e+08/3.6e+08	1.2e+09/1.5e+09	5.2e+05/1.6e+06	6.8e+04/1.4e+05	4.4e+03/2.0e+04	2.9e+03/2.4e+03	3.3e+06/2.4e+06

Table 41: Per-pass PAPI counters for `TernaryTree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
add 1 tree	M	3.2e+08 /1.6e+09	9.5e+08 /2.4e+09	2.4e+05 /2.3e+06	6.8e+04 /1.8e+05	3.3e+03 /4.4e+04	3.2e+03 /4.8e+03	1.5e+06/ 9.6e+05
sum tree	F	2.6e+08 /1.4e+09	5.8e+08 /1.0e+09	8.4e+04 /2.4e+06	9.8e+01 /1.1e+03	1.4e+03 /4.1e+04	9.9e+01 /3.0e+02	1.3e+06/ 9.0e+05
Total		5.8e+08/3.0e+09	1.5e+09/3.4e+09	3.2e+05/4.6e+06	6.8e+04/1.8e+05	4.8e+03/8.6e+04	3.3e+03/5.1e+03	2.8e+06/1.9e+06

Table 42: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
autocompleteTopKProxy	F	5/10	50%	0.0148 $\pm$ 0.00001	0.0270 $\pm$ 0.00089	0.0080$\pm$0.00003	0.1077 $\pm$ 0.00122	1.84$\times$	1.83$\times$	3.37$\times$	0.25 $\times$
countLazyNodes	F	4/10	60%	0.0126 $\pm$ 0.00000	0.0274 $\pm$ 0.00001	0.0065$\pm$0.00000	0.1086 $\pm$ 0.00002	1.93$\times$	2.16$\times$	4.19$\times$	0.25 $\times$
countTerminals	F	3/10	70%	0.0119 $\pm$ 0.00001	0.0254 $\pm$ 0.00001	0.0058$\pm$0.00000	0.0982 $\pm$ 0.00014	2.05$\times$	2.13$\times$	4.37$\times$	0.26 $\times$
decayTrieStats	M	10/10	0%	0.0741$\pm$0.00095	0.0794 $\pm$ 0.00009	0.0841 $\pm$ 0.00011	0.2470 $\pm$ 0.00127	0.88 $\times$	1.07 $\times$	0.94 $\times$	0.32 $\times$
resetTraversalState	M	10/10	0%	0.0748$\pm$0.00010	0.0806 $\pm$ 0.00009	0.0816 $\pm$ 0.00007	0.2339 $\pm$ 0.00115	0.92 $\times$	1.08 $\times$	0.99 $\times$	0.34 $\times$
sumPrefixFreq	F	3/10	70%	0.0120 $\pm$ 0.00001	0.0259 $\pm$ 0.00016	0.0057$\pm$0.00008	0.1048 $\pm$ 0.00003	2.10$\times$	2.15$\times$	4.50$\times$	0.25 $\times$
sumSubtreeHints	F	3/10	70%	0.0120 $\pm$ 0.00000	0.0258 $\pm$ 0.00001	0.0057$\pm$0.00000	0.1047 $\pm$ 0.00002	2.11$\times$	2.15$\times$	4.53$\times$	0.25 $\times$
Total				0.2123	0.2915	0.1975	1.005	1.07 $\times$	1.37$\times$	1.48$\times$	0.29 $\times$
Geomean								1.59$\times$	1.72$\times$	2.74$\times$	0.27 $\times$

Table 43: Per-pass PAPI counters for `Trie` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
autocompleteTopKProxy	F	7.5e+07/ 4.1e+07	1.3e+08 /2.1e+08	1.5e+06/ 4.9e+05	7.3e+01/ 3.9e+01	5.4e+04/ 4.7e+03	7.4e+01/ 4.0e+01	3.6e+06/ 7.5e+05
countLazyNodes	F	6.4e+07/ 3.3e+07	1.3e+08 /1.6e+08	1.4e+06/ 2.9e+04	1.0e+02/ 8.0e+01	1.6e+04/ 3.3e+03	9.8e+01/ 8.4e+01	4.0e+06/ 5.5e+05
countTerminals	F	6.0e+07/ 2.9e+07	1.0e+08 /1.2e+08	1.5e+06/ 1.1e+04	1.2e+02/ 6.6e+01	2.6e+04/ 2.3e+03	1.2e+02/ 6.7e+01	3.9e+06/ 2.1e+05
decayTrieStats	M	1.6e+08 /2.1e+08	3.5e+08 /7.9e+08	9.3e+05 /1.3e+06	8.8e+04 /1.1e+05	6.4e+03 /2.1e+04	2.2e+03 /2.4e+04	3.1e+06/ 1.2e+06
resetTraversalState	M	1.6e+08 /2.0e+08	2.8e+08 /7.0e+08	7.2e+05 /7.8e+05	9.5e+04 /1.0e+05	7.1e+03 /1.4e+04	2.2e+03 /1.6e+04	3.1e+06/ 9.1e+05
sumPrefixFreq	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.4e+04	9.1e+01/ 6.3e+01	2.9e+04/ 2.6e+03	8.8e+01/ 6.5e+01	3.9e+06/ 2.1e+05
sumSubtreeHints	F	6.1e+07/ 2.9e+07	1.1e+08 /1.2e+08	1.5e+06/ 1.4e+04	1.4e+02/ 6.5e+01	2.7e+04/ 2.5e+03	1.4e+02/ 6.6e+01	3.9e+06/ 2.0e+05
Total		6.5e+08/5.7e+08	1.2e+09/2.2e+09	8.9e+06/2.7e+06	1.8e+05/2.2e+05	1.7e+05/5.0e+04	4.9e+03/4.0e+04	2.6e+07/4.0e+06

Table 44: Per-pass PAPI counters for `Trie` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
autocompleteTopKProxy	F	1.4e+08 /5.5e+08	2.3e+08 /7.5e+08	1.1e+06 /1.5e+06	1.2e+02 /7.0e+02	3.0e+03 /2.7e+04	1.3e+02 /5.5e+02	2.8e+06/ 4.6e+05
countLazyNodes	F	1.4e+08 /5.5e+08	2.2e+08 /7.6e+08	2.2e+06/ 1.1e+06	6.5e+01 /6.8e+02	2.6e+03 /2.2e+04	6.8e+01 /2.4e+02	2.7e+06/ 3.2e+05
countTerminals	F	1.3e+08 /5.0e+08	2.0e+08 /7.3e+08	1.2e+06/ 6.4e+05	8.7e+01 /9.5e+02	1.6e+03 /1.6e+04	9.0e+01 /1.5e+02	2.8e+06/ 1.5e+05
decayTrieStats	M	1.9e+08 /1.0e+09	4.5e+08 /2.1e+09	3.3e+05 /5.3e+06	8.9e+04 /3.9e+05	3.6e+03 /3.4e+04	3.5e+03 /4.4e+04	3.0e+06/ 1.2e+06
resetTraversalState	M	2.0e+08 /9.7e+08	3.7e+08 /2.0e+09	3.1e+05 /3.4e+06	1.1e+05 /1.7e+05	4.4e+03 /2.7e+04	2.2e+03 /3.3e+04	2.9e+06/ 9.8e+05
sumPrefixFreq	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.2e+05	1.3e+02 /7.1e+02	1.3e+03 /1.6e+04	1.3e+02 /1.4e+02	2.8e+06/ 1.5e+05
sumSubtreeHints	F	1.3e+08 /5.3e+08	2.1e+08 /7.3e+08	1.4e+06/ 6.1e+05	7.1e+01 /7.7e+02	1.4e+03 /1.6e+04	7.1e+01 /1.1e+02	2.8e+06/ 1.4e+05
Total		1.1e+09/4.7e+09	1.9e+09/7.8e+09	7.8e+06/1.3e+07	2.0e+05/5.6e+05	1.8e+04/1.6e+05	6.3e+03/7.8e+04	2.0e+07/3.4e+06

Table 45: Per-pass performance for `ColorOmtree`, ADT has 25 fields, SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Si	Am/Sm	Ai/Am	Ai/Sm	Ai/Si
paletteEntriesQuantized	F	13/25	48%	0.0461 $\pm$ 0.00003	0.0524 $\pm$ 0.00028	0.0341$\pm$0.00087	<i>OOM</i>	1.35$\times$	1.14$\times$	1.54$\times$	–
quantizationErrorProxy	F	10/25	60%	0.0497 $\pm$ 0.00083	0.0636 $\pm$ 0.00001	0.0426$\pm$0.00001	<i>OOM</i>	1.17$\times$	1.28$\times$	1.49$\times$	–
Total				0.0958	0.1160	0.0767	–	1.25$\times$	1.21$\times$	1.51$\times$	–
Geomean								1.26$\times$	1.21$\times$	1.51$\times$	–

Table 46: Per-pass PAPI counters for `Color0ctree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (A/S)	INS (A/S)	L1D (A/S)	L1I (A/S)	L2D (A/S)	L2I (A/S)	LLC (A/S)
paletteEntriesQuantized	F	2.3e+08/ 1.7e+08	6.7e+08 /7.6e+08	5.3e+06/ 1.1e+06	4.2e+02/ 1.4e+02	1.2e+04 /4.0e+04	4.2e+02/ 1.5e+02	9.2e+06/ 1.4e+06
quantizationErrorProxy	F	2.5e+08/ 2.2e+08	9.7e+08 /1.2e+09	4.3e+06/ 1.9e+06	1.4e+02 /2.7e+02	1.5e+04 /3.2e+04	1.4e+02 /2.7e+02	9.1e+06/ 2.6e+06
Total		4.9e+08/3.9e+08	1.6e+09/2.0e+09	9.6e+06/3.1e+06	5.5e+02/4.1e+02	2.7e+04/7.3e+04	5.6e+02/4.1e+02	1.8e+07/4.0e+06

Table 47: Per-pass PAPI counters for `Color0ctree` (Ai/Si). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	T	CYC (Ai/Si)	INS (Ai/Si)	L1D (Ai/Si)	L1I (Ai/Si)	L2D (Ai/Si)	L2I (Ai/Si)	LLC (Ai/Si)
paletteEntriesQuantized	F	2.7e+08/−	7.8e+08/−	2.1e+06/−	2.8e+02/−	4.2e+03/−	2.8e+02/−	8.5e+06/−
quantizationErrorProxy	F	3.2e+08/−	1.1e+09/−	1.5e+06/−	4.5e+02/−	2.3e+04/−	4.5e+02/−	7.2e+06/−
Total		5.9e+08/−	1.8e+09/−	3.5e+06/−	7.3e+02/−	2.7e+04/−	7.3e+02/−	1.6e+07/−